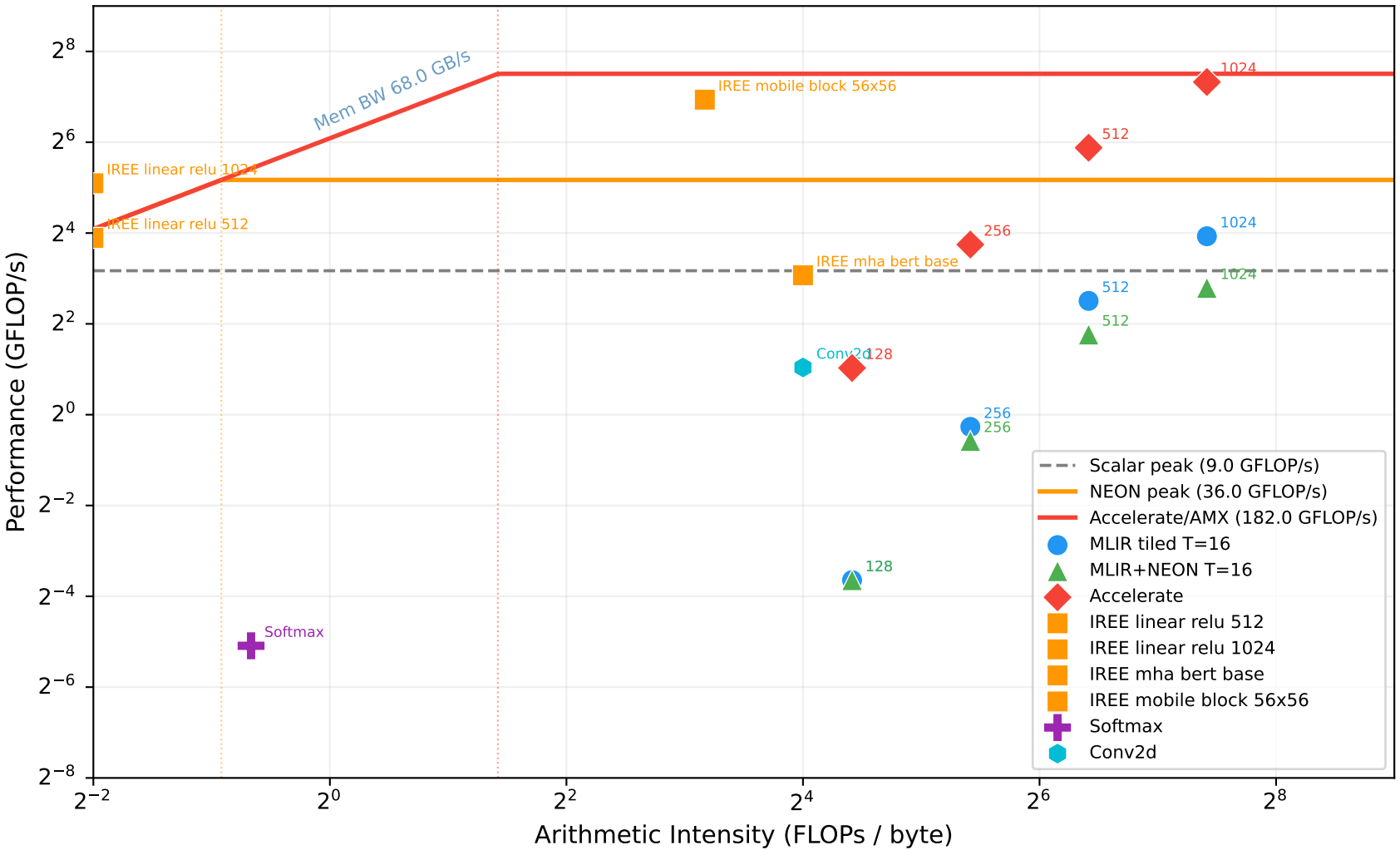
Roofline — Apple M4 Pro (single thread, FP32)
MLIR naive vs NEON vs Accelerate
Performance (GFLOP/s)
Arithmetic Intensity (FLOPs / byte)
Mem BW 68.0 GB/s
IREE linear relu 1024
IREE linear relu 512
IREE mobile block 56x56
IREE mha bert base
Conv2d
Softmax
1024
512
256
128
Scalar peak (9.0 GFLOP/s)
NEON peak (36.0 GFLOP/s)
Accelerate/AMX (182.0 GFLOP/s)
MLIR tiled T=16
MLIR+NEON T=16
Accelerate
IREE linear relu 512
IREE linear relu 1024
IREE mha bert base
IREE mobile block 56x56
Softmax
Conv2d